Chapter 1: Programming

Programming involves analyzing problems and developing logical sets of instructions (algorithms) to solve them. This chapter focuses on problem analysis, control structures, flowcharts, and Pascal programming.

1.1 Analyzing a Problem

Every problem-solving process consists of three main stages:

- **Input:** Raw materials or data needed to solve the problem.
- **Process:** The step-by-step sequence of actions taken.
- **Output:** The final result or information obtained.

1.2 Representation of Algorithms

An algorithm can be represented using **Flowcharts** (graphical) or **Pseudo-code** (structured text).

Flowchart Symbols

Symbol	Function
Oval (Terminator)	Start / End
Parallelogram	Input / Output
Rectangle	Process
Diamond	Decision (Yes/No)
Arrows	Flow Direction

1.3 Control Structures

- **Sequence:** Steps are executed one after another in order.
- **Selection (Condition):** Choosing a path based on a condition (e.g., IF-THEN-ELSE).
- **Iteration (Repetition):** Repeating a set of steps multiple times (e.g., FOR, WHILE, REPEAT-UNTIL).

1.4 Programming in Pascal

Pascal is a structured programming language. A basic program structure includes:

```
program ProgramName;
var
  VariableList : DataType;
begin
  { Instructions }
  writeln('Hello World');
end.
```

Common Pascal Data Types

- **Integer:** Whole numbers.
- **Real:** Decimal numbers.
- **Char:** A single character.
- **String:** A sequence of characters.
- **Boolean:** True or False values.

1.5 Evolution of Languages

- **1st Generation:** Machine Language (0s and 1s).
- **2nd Generation:** Assembly Language (using mnemonics).
- **3rd Generation:** High-Level Languages (Pascal, C, Java).
- **4th Generation:** Human-like languages (SQL).
- **5th Generation:** Based on Artificial Intelligence (Prolog, LISP).